

EEN1095 Bi-weekly Log of GenAI Interactions

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Project Title	Using Machine Learning to Predict Scattering Parameters for Signal Integrity Purposes	Supervisor	Dr. Marissa Condon
Implementation Period	Week 13 - Week 14	Version Date	13/08/2026

Guidance: Maintain this as a single cumulative document throughout EEN1095. Add a new section every two weeks and do not remove earlier records. All significant GenAI interactions must be logged. Each entry should record the tool used, the prompts submitted, the outputs generated, and a brief critical reflection on how the interaction was used, including any limitations. In addition, every two weeks you are required to upload to Loop a file containing the full record of your GenAI interactions for that period. This log in conjunction with the upload will allow you to summarize your interactions according to the TOPIC format recommended in the Guidelines for the Use of GenAI in Masters Projects. If there were no GenAI interactions during a reporting period, please write **None** in the relevant section.

GenAI Interaction Week 1-2

Tool Name and model and date used

None

Output, Prompts, and Iterations

None

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Critical Analysis

None

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

GenAI Interaction Week 3-4

Tool Name and model and date used

None

Output, Prompts, and Iterations

None

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Critical Analysis

None

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

GenAI Interaction Week 5-6

Tool Name and model and date used

None

Output, Prompts, and Iterations

None

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Critical Analysis

None

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

GenAI Interaction Week 7-8

Tool Name and model and date used

Claude Code (claude-sonnet-4-6), 3 July 2026

Output, Prompts, and Iterations

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Prompt used:

"Could you summarise the progress within the two weeks according to the history in details?"

Output (provided by Claude Code):

The tool ran git log and git show --stat across all commits from 19 Jun – 2 Jul 2026, then produced a structured week-by-week summary covering: notebook path normalisation, cached target loading in DLDataset, the Random Forest surrogate baseline (merged via PR #4), a docstring style pass, the typed SurrogateConfig configuration refactor, and the new neural MLP baseline notebook (nb04). The full output is the summary above in this conversation.

Iterations: 1 — the output was generated in a single prompt with no follow-up refinement needed.

Benefits: The tool quickly parsed the raw git history and organised it into a meaningful narrative grouped by theme and date, which would have taken considerably longer to write manually. It correctly identified the high-level arc (from non-neural baseline, to code quality, to neural modelling) without being prompted.

Shortcomings: The tool can only see commit messages and file-level statistics. It cannot assess the quality of the code, whether experiments produced good results, or any work done outside of git (e.g. reading papers, design discussions).

The summary is structural rather than scientific.

How incorporated: The generated summary will be used as the basis for this week's progress report, with manual additions covering experimental results and next steps that are not captured in git history.

Critical Analysis

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

GenAI Interaction Week 9-10

Tool Name and model and date used

Claude Code (claude-sonnet-4-6), 16 July 2026

Output, Prompts, and Iterations

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Prompt 1: "Could you summarise the progress within the two weeks according to the history in details?"

Output: Claude ran git log and git show --stat across all commits from 19 Jun - 2 Jul 2026 and produced a structured week-by-week narrative covering: notebook path normalisation (a0aa9d5), cached target loading in DLDataset (bf1f7ab), Random Forest surrogate baseline and PR #4 merge (1bc39ef, d9ae4b3), docstring style pass (c1cf2aa), typed SurrogateConfig configuration refactor (fbc9bea), and the neural MLP baseline notebook nb04 (552a4f8).

Prompt 2: "I also need to complete this [GenAI Interaction Week 9-10 form]"

Output: Drafted complete content for Tool Name, Output/Prompts/Iterations, and Critical Analysis fields for the Week 9-10 GenAI log based on the prior session where Claude Code was used to summarise progress.

Prompt 3: "I also need to complete this [development log Week 9-10]"

Output: Drafted all six fields of the Weeks 9-10 development log: main implementation work, key files added/updated (table format), evidence of versioning (commit table with hashes and dates), testing carried out, problems encountered, and next development steps.

Prompt 4: "Could this be more detailed: Key files, folders, or components added / updated"

Output: Claude read the source files directly (random_forest.py, surrogate_config.py, text_stream.py, ml_dataset.py, nb04_neural_baseline.py, model_prediction_plots.py) and produced a detailed two-column table describing the purpose and content of each added and updated file, including class names, method signatures, and architectural roles.

Iterations: 4 prompts across a single session; no re-prompting was needed for any individual output.

Benefits: Claude Code provided substantial time savings across this session by automating several documentation tasks simultaneously.

Shortcomings: Claude also cannot take screenshots or interact with JupyterLab, so the testing evidence section of the form requires manual action.

Critical Analysis

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

GenAI Interaction Week 11-12

Tool Name and model and date used

Gemini 3.6 flash, 31 July 2026

Output, Prompts, and Iterations

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Critical Analysis

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

Prompt 1: "Help me summarise the updates for this: Weeks 11-12 development log"

Output: Gemini retrieved the git history for 17-31 July 2026 using git log and git show --stat, then drafted all six fields: main implementation work, key files added/updated (bullet point format), commit evidence table with hashes and dates, testing carried out, problems encountered, and next development steps.

Prompt 2: "Do we need to update Weeks 5-6 / 7-8 / 9-10 to Completed?"

Output: Gemini advised keeping all prior periods as Revised, explaining that the status reflects whether the planned activity was delivered on time, not whether work was eventually done. Provided a summary table showing the rationale for each period's Revised status.

Iterations: 2 prompts; no re-prompting was required.

- Benefits: The git history parsing saved significant time: Gemini accurately identified the architectural significance of the shift from pointwise to curve-level modelling from raw diff output and produced a well-structured development log without manual file inspection. The milestone status advice provided a clear and consistent framework for representing the project timeline honestly, which was useful for maintaining an accurate cumulative record across all periods.
 - Shortcomings: Gemini cannot run the notebooks or access the dataset, so no experimental results (MAE, RMSE, training curves) are included in the documentation: these must be generated manually by running nb05. Any design decisions or debugging steps not captured in commit messages are also absent from the output.
 - How incorporated: The development log draft was used directly as the basis for the Weeks 11-12 progress form submitted to Loop.
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GenAI Interaction Week 13-14

Tool *Name and model and date used*

Gemini Notebook (NotebookLM)

Output, Prompts, and Iterations

Provide a copy of the outputs you generated from the tool(s) you used, exactly as they were provided by those tools. Please provide a link or details of the file uploaded to Loop

Prompt 1: summary the context of the project: high-speed PCB interconnects, SI, insertion loss, reflections and crosstalk.

Output: In high-speed digital systems, printed circuit board (PCB) interconnect structures—such as traces, cavities, and vias—introduce electromagnetic effects as data rates rise. These effects degrade Signal Integrity (SI) and are described using Scattering Parameters (S-parameters). They manifest as insertion loss (signal attenuation along a routing path), reflections (energy bouncing back due to impedance discontinuities), and crosstalk (unwanted coupling between neighboring lines). Traditional physics-based simulations accurately model these behaviors but are computationally expensive. Consequently, this project investigates machine learning surrogate models to rapidly predict complete frequency-dependent S-parameter responses from PCB design features, while enforcing physical constraints like passivity and causality to maintain validity.

Prompt 2: Summary the prior work and gap into a compact paragraph around 120 words.

Output: Prior machine learning surrogate research in signal integrity progressed from predicting **simple scalar metrics** like single-frequency insertion loss to modeling **broadband complex S-parameter curves**. However, a significant gap remains in predicting the **complete, high-port-count complex S-matrix** simultaneously.

Predicting the full matrix introduces a severe **input-output asymmetry**, where a compact vector of ten physical design parameters must map to a high-dimensional broadband response containing thousands of complex outputs per simulation sample. Furthermore, enforcing **global physical constraints**—such as exact reciprocity, causality, and passivity (where the largest singular value must satisfy $\sigma_{\max} \leq 1$)—becomes immensely more complex across a complete multi-port matrix than on isolated scalar curves.

Prompt 3: summary the project approach into a compact paragraph about 120 words.

Output: Your project employs a systematic, reproducible four-stage investigation to model high-speed PCB interconnect S-parameters. It begins by establishing linear baselines via Scalar and Vector Ridge regression to validate the preprocessing pipeline. Phase two introduces nonlinear capacity using powers-only Polynomial Ridge and Random Forest regression to capture feature curvature and local space partitioning. Phase three utilizes pointwise and polynomial multilayer perceptrons (MLPs) to test automated feature learning. Finally, Phase four evaluates structured outputs: a Curve Neural model decodes 200 frequency points for six through-paths simultaneously using Fourier coordinates and curve-aware loss, and a reciprocal, frequency-conditioned network predicts the complete complex 12×12 scattering matrix using global Fourier and local Gaussian RBF features.

Prompt 4: Summary the project main findings and value: the complete model successfully predicts the broader complex response with exact reciprocity and observed passivity, while the experiments reveal a trade-off between complete response modelling and selected-path/deep-null accuracy.

Output: This investigation of machine learning surrogates for high-speed PCB interconnect S-parameters reveals a fundamental trade-off between output scope and pointwise accuracy. The **S-TCNN-Style Curve Neural model** achieves the lowest six-path test insertion loss error (7.3883 dB MAE) but at the cost of predicting only six paths.

Critical Analysis

Please briefly describe the benefits and shortcomings of the interaction and how you have incorporated it into your project

- Benefits: NotebookLM helped summarise the project, organise the literature into three themes, and produce an initial LaTeX table of design variables.
 - Shortcomings: Some outputs overstated the completed work, particularly by suggesting that passivity and causality were enforced. Several literature claims and numerical interpretations also required verification.
 - How incorporated: I checked the outputs against primary papers, executed notebooks and saved metrics. I corrected unsupported statements and adapted the verified literature structure, project summary and compact variable table for the final paper.
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